MSG No. 182.180 $P6_322'$ [Type II, hexagonal]

Table 1: Wyckoff site: 2a, site symmetry: 32.1'

No.	position	mapping
1	$[0, 0, 0]$	[1,3,5,7,8,9,13,15,17,19,20,21]
2	$\left[0, 0, \frac{1}{2}\right]$	[2,4,6,10,11,12,14,16,18,22,23,24]

Table 2: Wyckoff site: 2b, site symmetry: 3.21'

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	[1,3,5,10,11,12,13,15,17,22,23,24]
2	$\left[0, 0, \frac{3}{4}\right]$	[2,4,6,7,8,9,14,16,18,19,20,21]

Table 3: Wyckoff site: 2c, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	[1,3,5,10,11,12,13,15,17,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	[2,4,6,7,8,9,14,16,18,19,20,21]

Table 4: Wyckoff site: 2d, site symmetry: 3.21'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	[1,3,5,10,11,12,13,15,17,22,23,24]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	[2,4,6,7,8,9,14,16,18,19,20,21]

Table 5: Wyckoff site: 4e, site symmetry: 3..1'

No.	position	mapping
1	$[0, 0, z]$	[1,3,5,13,15,17]
2	$\left[0, 0, z + \frac{1}{2}\right]$	[2,4,6,14,16,18]
3	$[0, 0, -z]$	[7,8,9,19,20,21]
4	$\left[0, 0, \frac{1}{2} - z\right]$	[10,11,12,22,23,24]

Table 6: Wyckoff site: 4f, site symmetry: 3..1'

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,3,5,13,15,17]
2	$\left[\frac{2}{3}, \frac{1}{3}, z + \frac{1}{2}\right]$	[2,4,6,14,16,18]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9,19,20,21]
4	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2} - z\right]$	[10,11,12,22,23,24]

Table 7: Wyckoff site: `6g`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7,13,19]`
2	$\left[x,\ x,\ \frac{1}{2}\right]$	`[2,10,14,22]`
3	$[0,\ x,\ 0]$	`[3,8,15,20]`
4	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[4,11,16,23]`
5	$[-x,\ -x,\ 0]$	`[5,9,17,21]`
6	$\left[0,\ -x,\ \frac{1}{2}\right]$	`[6,12,18,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `..21'`

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{4}\right]$	`[1,11,13,23]`
2	$\left[-x,\ x,\ \frac{3}{4}\right]$	`[2,9,14,21]`
3	$\left[-2x,\ -x,\ \frac{1}{4}\right]$	`[3,12,15,24]`
4	$\left[-x,\ -2x,\ \frac{3}{4}\right]$	`[4,7,16,19]`
5	$\left[x,\ -x,\ \frac{1}{4}\right]$	`[5,10,17,22]`
6	$\left[2x,\ x,\ \frac{3}{4}\right]$	`[6,8,18,20]`

Table 9: Wyckoff site: `12i`, site symmetry: `11'`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1,13]`
2	$\left[x-y,\ x,\ z+\frac{1}{2}\right]$	`[2,14]`
3	$[-y,\ x-y,\ z]$	`[3,15]`
4	$\left[-x,\ -y,\ z+\frac{1}{2}\right]$	`[4,16]`
5	$[-x+y,\ -x,\ z]$	`[5,17]`
6	$\left[y,\ -x+y,\ z+\frac{1}{2}\right]$	`[6,18]`
7	$[x-y,\ -y,\ -z]$	`[7,19]`
8	$[y,\ x,\ -z]$	`[8,20]`
9	$[-x,\ -x+y,\ -z]$	`[9,21]`
10	$\left[x,\ x-y,\ \frac{1}{2}-z\right]$	`[10,22]`
11	$\left[-x+y,\ y,\ \frac{1}{2}-z\right]$	`[11,23]`
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	`[12,24]`